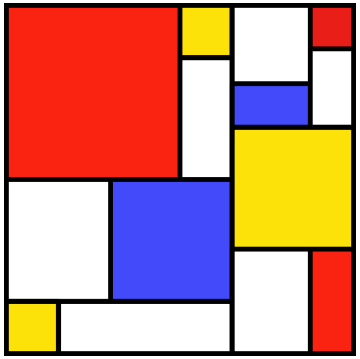
SLiM

Workshop Series



Recombination

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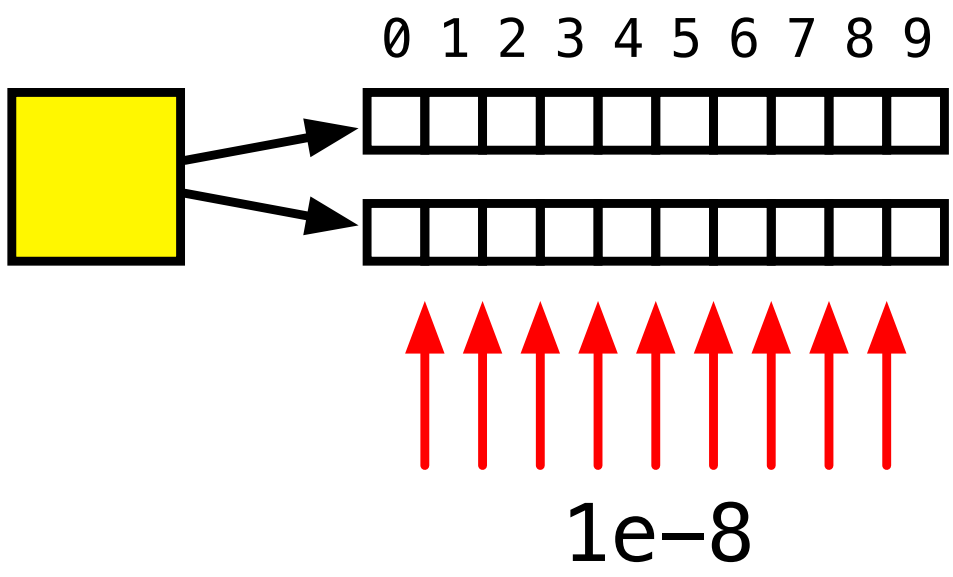
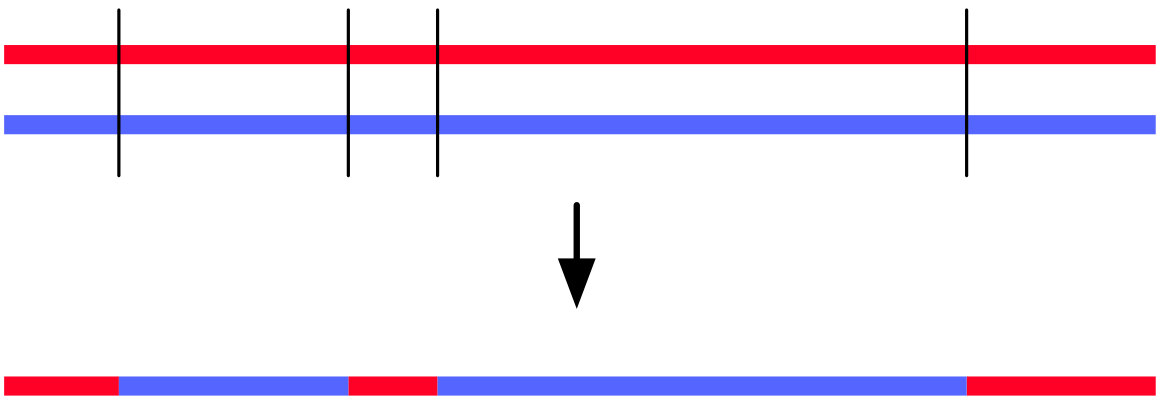
Recombination

- Can be given a uniform rate:
  - `initializeRecombinationRate(1e-8)`
- *Probability* / base / generation



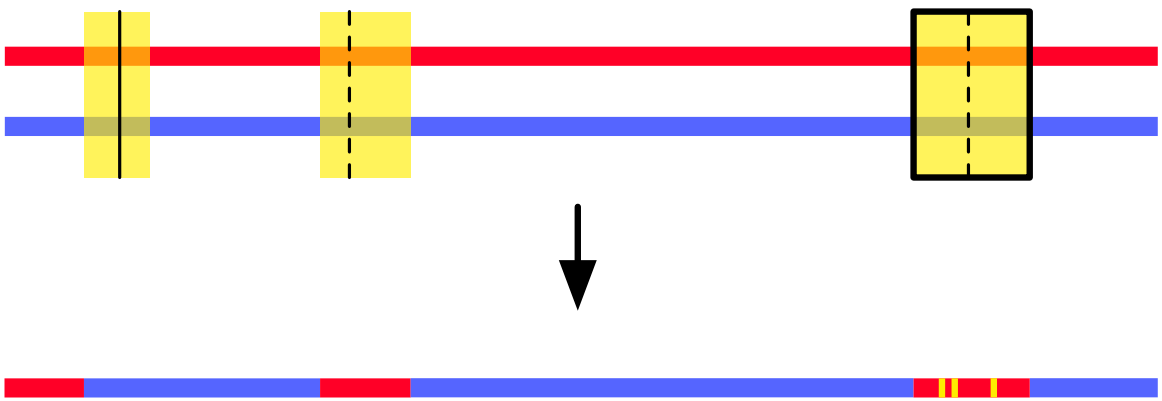
The Crossover Breakpoints Model

- During gamete generation:
  - breakpoints are drawn by probability
  - a simple **crossover model** is the default



The DSB (double-stranded break) Model

- Each breakpoint initiates *gene conversion*
  - DSBs can be **crossovers** or **non-crossovers**
  - gene conversion tracts can be **simple** or **complex**
  - heteroduplex mismatch repair, GC biased repair



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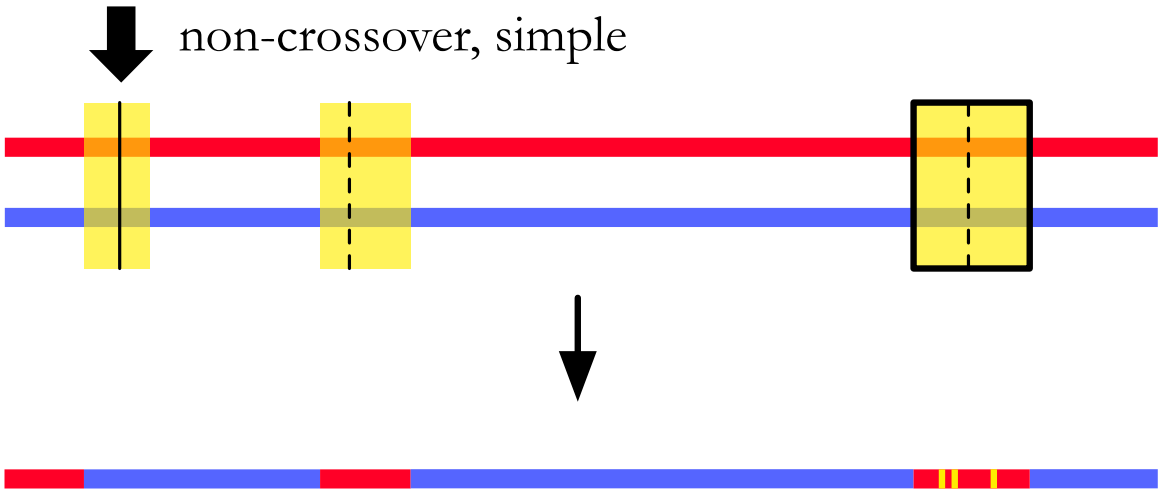

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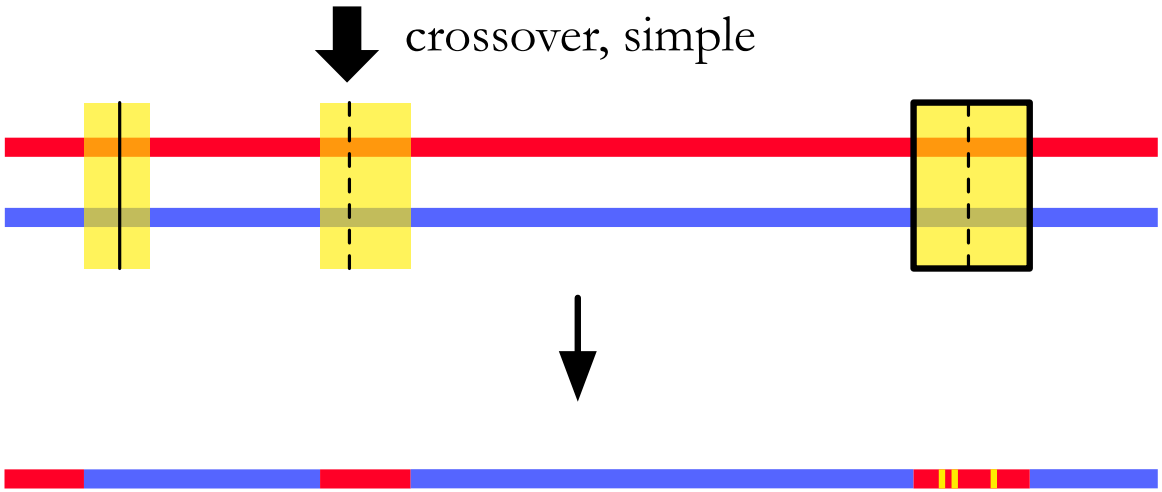

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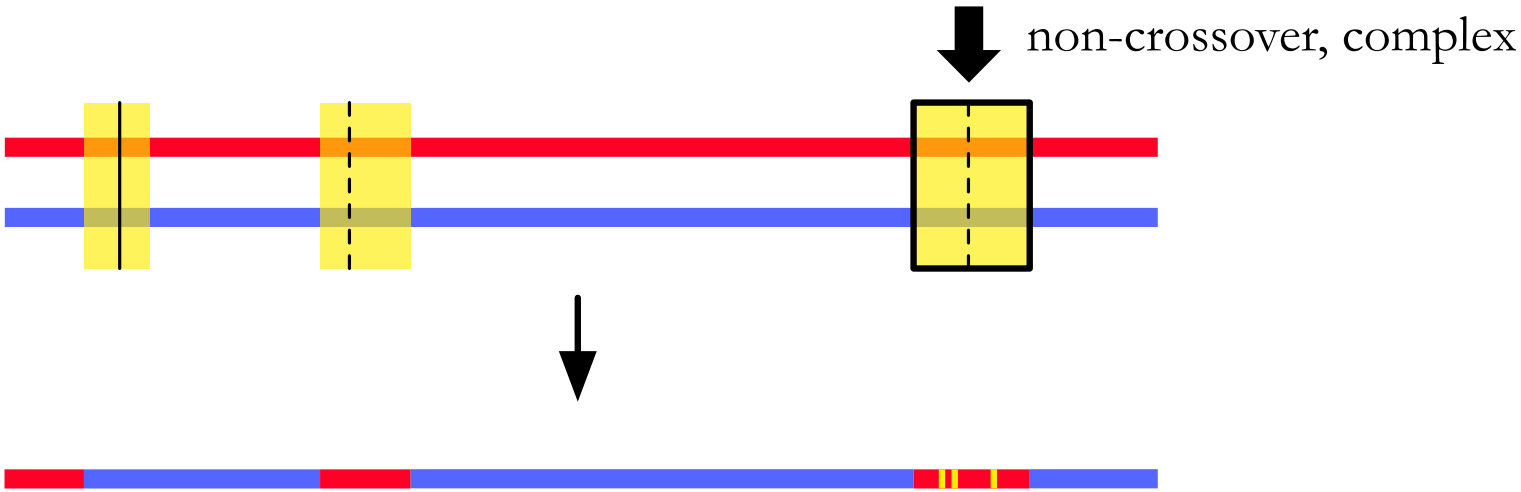

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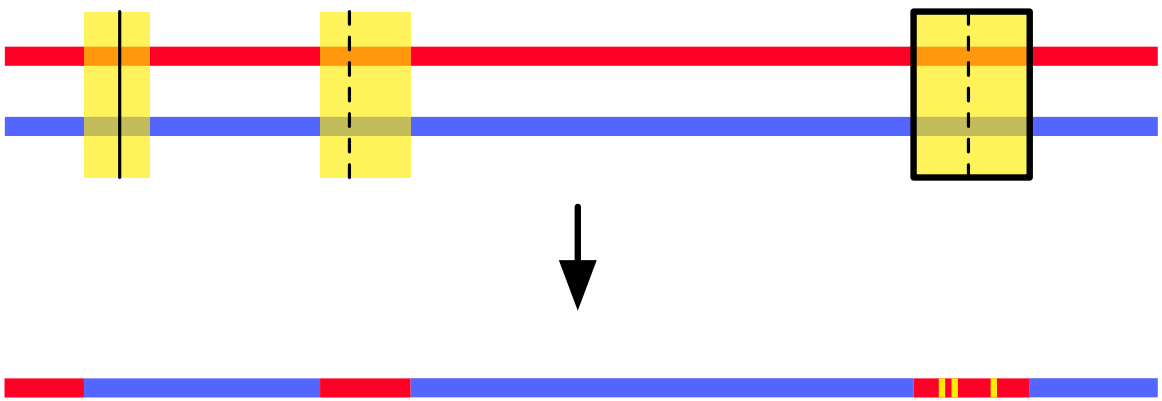


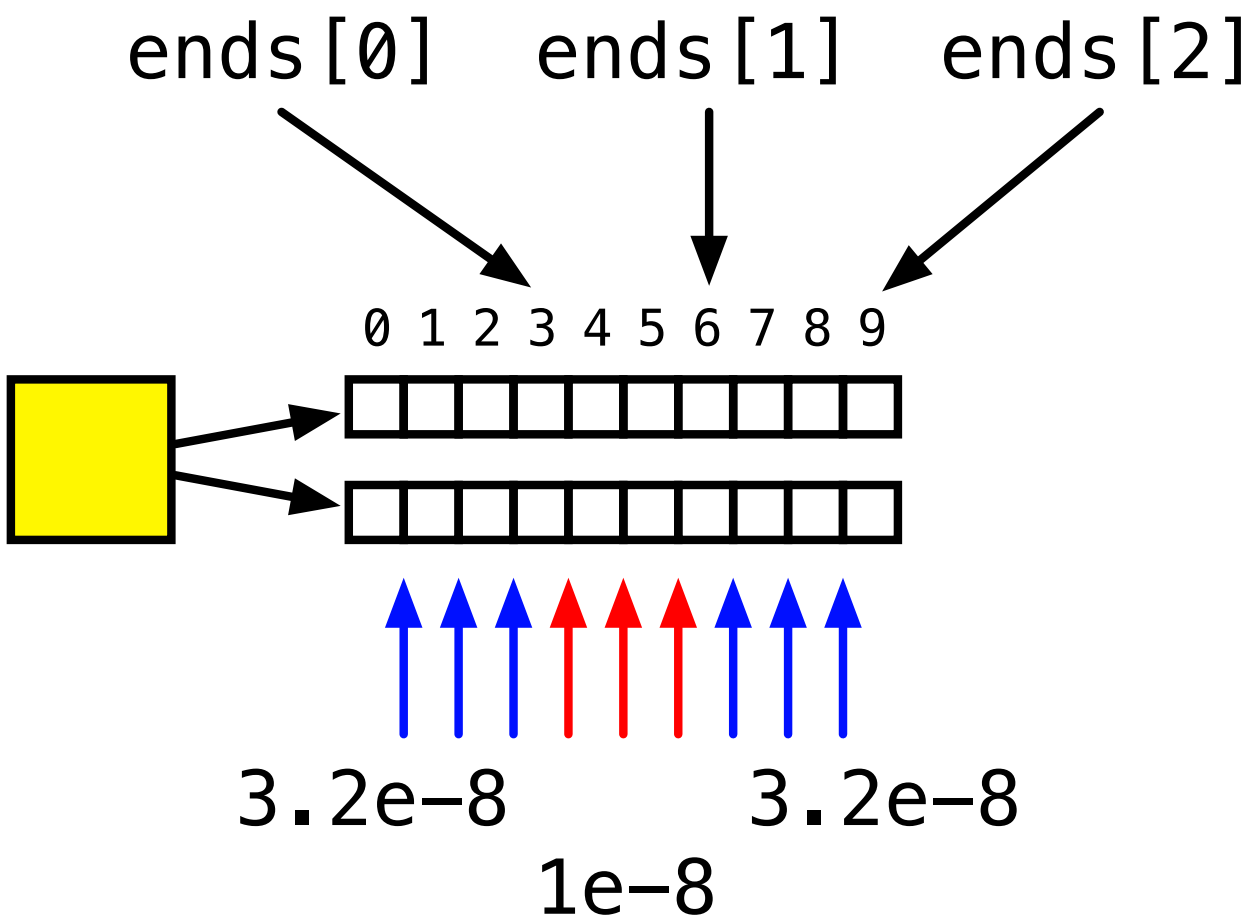
Recombination Rate Maps

- Can also be given vectors of *rates* and *ends*:

```
(void)initializeRecombinationRate(numeric rates,
    [Ni ends = NULL], [string$ sex = "*"])
```

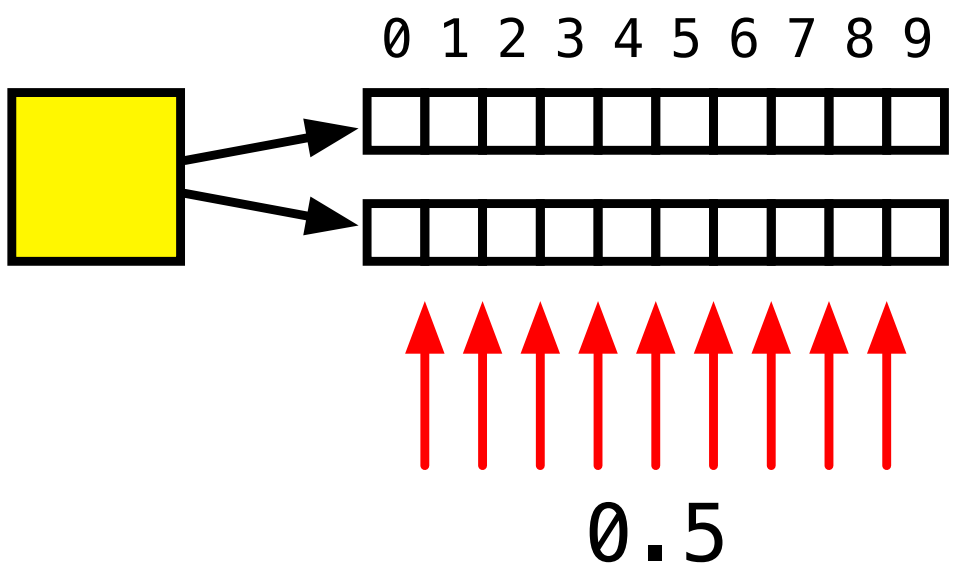
- Rates apply until the **left edge** of each end

```
initializeRecombinationRate(
   rates = c(3.2e-8, 1e-8, 3.2e-8),
   ends = c(3, 6, 9));
```



Unlinked Loci

- A rate of **0.5** makes regions **unlinked**
- Model any number of unlinked loci



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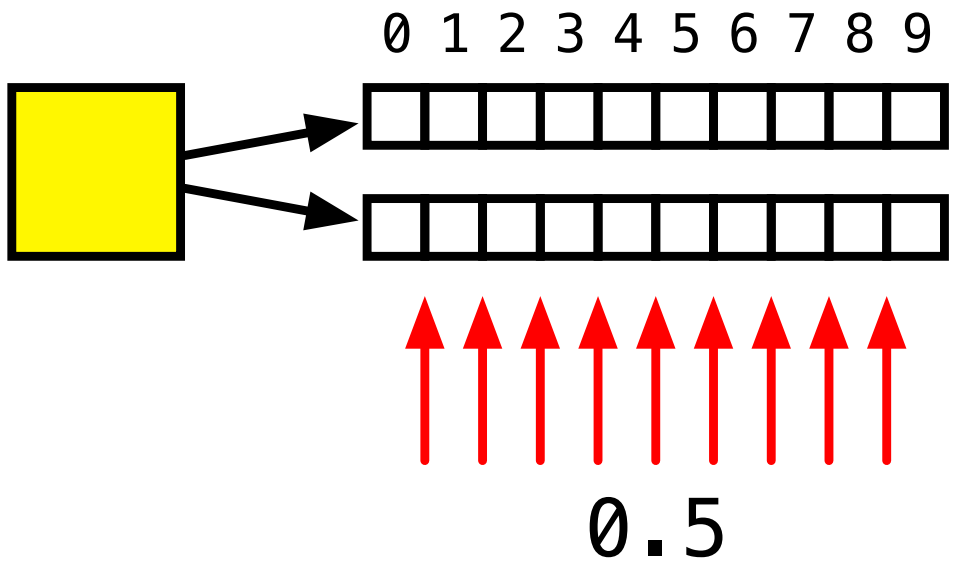
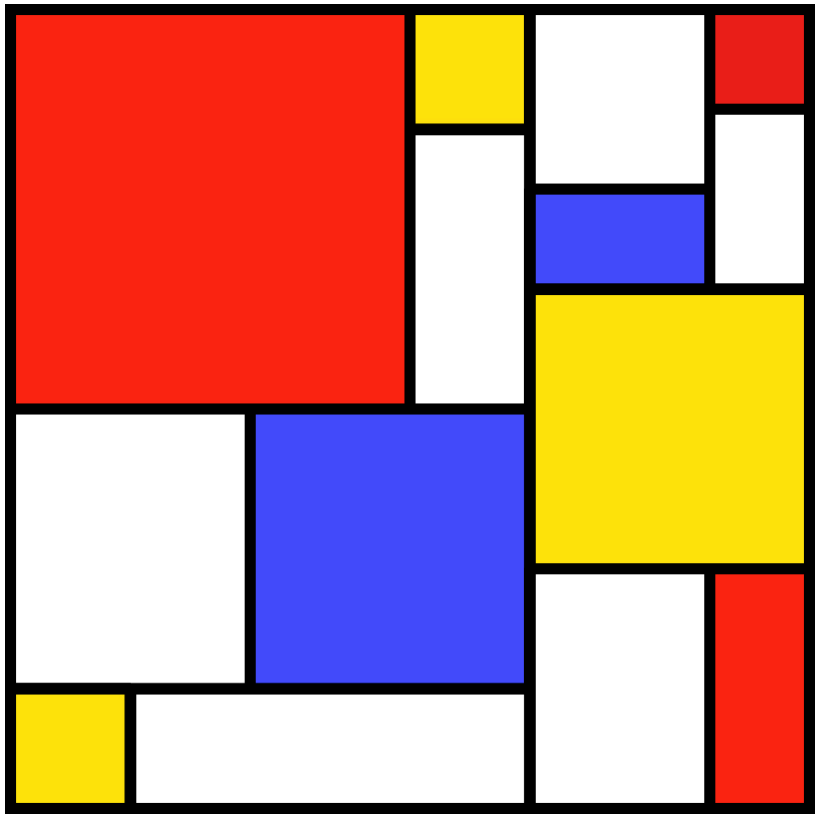


- Performance can be an issue!

Multiple Chromosomes

- SLiM 5 supports multiple chromosomes
- Usually loci are unlinked because they're on different chromosomes!
- We will cover multiple chromosomes later in the workshop, stay tuned



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